

Lecture 16

DATA 322: Machine Learning for Data Science

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Learning objectives (Chapters 1 and 2)

1. Distinguish between supervised and unsupervised learning.
 2. Distinguish among descriptive, predictive, estimation, and testing analyses; use random-variable vs. summary notation; and recognize confounding and control for bias in estimation and testing.
 3. Use appropriate notation to describe a supervised learning problem (available dataset, hypothetical dataset, true relationship, predicted relationship, predicted target, training data, test data).
 4. State the bias–variance decomposition of the expected test error.
 5. Classify a supervised learning problem as regression or classification.
 6. Explain flexibility of a model and its role in the bias–variance trade-off.
 7. Explain interpretability of a model and when it matters.
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1. Supervised vs unsupervised learning

Supervised learning: Imagine we have a *response* (or *outcome*, *target*) variable Y that we want to predict. Imagine we have one or more *features* (predictors or inputs) $X_1, \dots, X_p$ that might provide information to improve the prediction. The goal of **supervised learning** is to learn a function that maps the features to the response, using a *labeled* dataset where values of Y are recorded along with values of the features for a set of examples. Two main tasks are **regression** (when Y is quantitative) and **classification** (when Y is categorical).

Unsupervised learning: We observe only features $X_1, \dots, X_p$; there is no response variable. The goal is to find structure in the data, e.g., groups (clustering), low-dimensional summaries, or associations—without a target to predict.

Example 1: Predict building-level energy use (regression) or whether a site is suitable for solar (classification) from building and location features. Unsupervised: cluster neighborhoods by energy use, renewables, and transport to design targeted sustainability programs.

Example 2: Predict next quarter's sales (regression) or whether a customer will churn (classification) from purchase history and demographics. Unsupervised: segment customers by spending, frequency, and product mix for targeted marketing.

Example 3: Predict weekly candy sales per store (regression) or whether a new product will be a top seller (classification) from price, shelf position, and store type. Unsupervised: cluster candy SKUs by sugar content, size, and price point to design product lines or store layouts.

Example 4: Predict length of stay or readmission risk (regression or classification) from vitals, labs, and demographics. Unsupervised: cluster patients by diagnoses, procedures, and utilization to identify care pathways or risk strata.

2. Types of data analysis: descriptive, predictive, estimation, testing

Data analyses can be classified into four types. Using **random variable** vs. **summary** notation helps keep the goal clear: we use capital letters (e.g. X , Y) for random variables and symbols like $\bar{x}$, $\hat{\beta}$, or population parameters μ , β for summaries or quantities we estimate/test.

Descriptive (what's in the dataset I have): Summarize the observed dataset only with no formal notion of a

larger population or future individuals. We describe the data at hand.

Predictive (a guess for an individual's value): For a *specific* unit (e.g., a new customer, next week's total sales, a new patient), we want to predict that unit's outcome. *Notation:* for a new or future unit we have inputs X^0 and want to predict the random variable Y^0 ; our guess is $\hat{Y}^0 = \hat{f}(X^0)$. The focus is on accuracy of the predicted value for that unit: mere correlation between inputs and output is fine.

Estimation (What is the summary of a population or relationship?): We want to know summary that describes a *population* or a *relationship* (e.g., population mean μ , average treatment effect) but have only data from a subset of the population. *Notation:* population quantities are parameters/quantities that are fixed (as opposed to random variables) (e.g. μ , β); we can estimate them with sample-based estimators $\hat{\mu}$, $\hat{\beta}$ and report confidence intervals. If we are estimating a relationship or treatment effect, **confounding** can bias our estimate (this is when a third variable influences both the predictor (or treatment) and the outcome) so the raw association does not equal the causal effect. We can **control for confounders** (e.g., by "adjusting" for them, or by study design such as randomized control trials) so that the estimated relationship is less biased. Causal diagrams (DAGs) help identify which variables to condition on: we typically need to adjust for variables that cause both treatment and outcome (backdoor paths) to avoid bias.

Testing (is the summary or relationship in the population different from...?): We ask whether a population quantity or relationship differs from a benchmark (e.g., zero, a null value) based on data from a subset of the population. *Notation:* we state a null hypothesis about a parameter (e.g. $H_0 : \beta = 0$) and use the sample to compute a test statistic and p -value. **Causal/confounding:** The same confounding issues as in estimation apply: if we test "does X affect Y ?" without controlling for confounders, the test can be misleading (e.g., significant only because of confounding, or non-significant because we did not control and the effect is masked). Controlling for bias via appropriate adjustment or design is part of valid testing when the goal is to learn about a population relationship or causal effect.

Example 1: Describe average emissions and correlation with income in the sample (descriptive). Predict next week's demand for the grid (predictive). Estimate how much rooftop solar reduces emissions in the population, controlling for income and region to limit confounding (estimation). Test whether the population effect of solar on emissions is different from zero (testing).

Example 2: Describe mean sales and distribution by region in the data (descriptive). Predict next month's demand for a specific store (predictive). Estimate how much a 10% price increase reduces unit sales in the population, controlling for season and store type (estimation). Test whether that effect differs from zero (testing).

Example 3: Describe candy sales and price in the sample (descriptive). Predict Halloween demand by store (predictive). Estimate how much shelf placement affects sales in the population, controlling for store size and promotion (estimation). Test whether the effect of placement is zero (testing).

Example 4: Describe readmission rate and lab values in the sample (descriptive). Predict which patient will need ICU in 24 hours (predictive). Estimate how much a care bundle reduces readmission in the population, controlling for severity and comorbidities to limit confounding (estimation). Test whether the treatment effect is different from zero (testing).

3. Notation for a supervised learning problem

In ISL style:

- **Available (training) data:** n examples $\{(x_1, y_1), \dots, (x_n, y_n)\}$, often written as design matrix X and response vector y . Together these form the training dataset S . More than one feature can be used, in which case each x_i is a vector of length p
- **True relationship:** $Y = f(X) + \varepsilon$, where f is the unknown true function and ε is irreducible error.
- **Estimated (predicted) relationship:** $\hat{f}(X)$, learned from the training data.
- **Predicted value:** $\hat{Y} = \hat{f}(X)$ for a given X (training or new).
- **Hypothetical (test) data:** New pairs (X^0, Y^0) not used to fit $\hat{f}$; used to evaluate prediction error, e.g. $E[(Y^0 -$

$$\hat{f}(X^0))^2].$$

Example 1: Y = annual CO₂ emissions per capita; X = GDP per capita, energy mix, population density, policy indices. We fit $\hat{f}$ on country-year data; the “hypothetical” set might be future years or held-out countries to assess generalizability of emission predictions.

Example 2: Y = revenue per customer; X = tenure, product count, support tickets, region. We fit $\hat{f}$ on historical accounts; the hypothetical set might be next quarter or a held-out region to evaluate revenue forecasts.

Example 3: Y = candy sales (units or revenue) per store-week; X = price, shelf position, season, store size, promotion. We fit $\hat{f}$ on past store-weeks; the hypothetical set might be next quarter or held-out stores to evaluate demand forecasts.

Example 4: Y = readmission within 30 days or blood glucose level; X = labs, vitals, medications, demographics, prior visits. We fit $\hat{f}$ on historical patients; the hypothetical set might be future admissions or held-out hospitals to evaluate risk or outcome predictions.

4. Bias–variance decomposition of expected test error

For squared-error loss and a **fixed** test input X^0 , the **expected test error** (expectation over training sets) is:

$$\mathbb{E}[(Y^0 - \hat{f}(X^0))^2] = \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible}} + \underbrace{[\mathbb{E}[\hat{f}(X^0)] - f(X^0)]^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(X^0))}_{\text{Variance}}$$

If X^0 is **random** (e.g., drawn from the same distribution as future inputs), we take expectation over both the training set and X^0 . That adds an outer $\mathbb{E}_{X^0}$: the expected test error becomes $\mathbb{E}_{X^0}[\mathbb{E}[(Y^0 - \hat{f}(X^0))^2 | X^0]]$, and the same decomposition holds *given* X^0 , so we just average the right-hand side over the distribution of X^0 .

- **Irreducible error:** Noise in Y that cannot be reduced by better modeling.
- **Bias:** Systematic error from approximating f by a simpler or restricted $\hat{f}$; often high when the model is too rigid (underfitting).
- **Variance:** Variability of $\hat{f}$ across different training samples; often high when the model is too flexible (overfitting).

Example 1: Predicting crop yield under climate change: a very simple model (bias-dominated) may miss nonlinear effects of temperature and water; a very complex model (variance-dominated) may overfit to a few regions.

Example 2: Predicting candy demand: a simple model may miss Halloween and holiday spikes; an overly flexible model may overfit to one year’s promotions and not generalize well to future promotions.

Example 3: Predicting readmission risk: a simple model may miss interactions between conditions and social determinants; an overly flexible model may overfit to one hospital’s coding or population.

5. Regression vs classification

Regression: The response Y is **quantitative** (continuous or discrete numeric). We predict a number (e.g., energy use, temperature, emissions). Typical loss: squared error; metrics: RMSE, MAE, R^2 .

Classification: The response Y is **categorical** (e.g., yes/no, land-use type). We predict a class. Typical loss: 0–1 loss or cross-entropy; metrics: accuracy, precision, recall, AUC.

Example 1: Regression: predict tons of CO₂ sequestered per hectare from forest type, age, and climate. Classification: classify satellite imagery into forest / agriculture / urban to monitor deforestation or urban sprawl.

Example 2: Regression: predict daily demand or delivery time from season, promotions, and store features. Classification: classify leads as likely-to-convert vs not, or transactions as fraudulent vs legitimate.

Example 3: Regression: predict sugar content (g per serving) or sales from ingredients, price, and package size. Classification: classify candy as chocolate / fruit / nut / other, or as “kid-focused” vs “adult” for shelf placement.

Example 4: Regression: predict length of stay or blood glucose from vitals, labs, and medications. Classification: classify patients as high-risk vs low-risk for readmission, or diagnose disease category from symptoms and lab results.

6. Flexibility of a model

Flexibility is the capacity of a model to fit different shapes of the input–output relationship. **Low flexibility** (e.g., linear model): smooth, stable fits, but may miss curvature and interactions (high bias, low variance). **High flexibility** (e.g., deep trees, many predictors): can fit complex and nonlinear patterns, but may overfit (low bias, high variance). Tuning (e.g., regularization, tree depth) controls flexibility and the bias–variance trade-off.

Example 1: Modeling species presence from climate and land use: a linear model may be too rigid; a flexible model can capture thresholds and interactions (e.g., temperature $\times$ rainfall) but may not generalize well.

Example 2: Modeling conversion rate from ad spend and demographics: a linear model may miss saturation effects and segment-specific response; a very flexible model may overfit to short campaigns.

Example 3: Modeling candy preference or sales from sugar content and type: a linear model may miss a “sweet spot” (preference peaks at moderate sugar); a very flexible model may overfit to a few brands.

Example 4: Modeling treatment response or disease progression from biomarkers and demographics: a linear model may miss dose–response thresholds or subgroup effects; a very flexible model may overfit to a single site or cohort.

7. Interpretability of a model

Interpretability is the ease with which a human can understand *how* the model uses inputs to form predictions or conclusions. **Interpretable** models (e.g., linear regression, small trees, additive models) allow one to inspect coefficients, variable importance, or simple rules. **Black-box** models (e.g., large ensembles, deep nets) may be more accurate but harder to explain.

Example 1: An interpretable model might show: “each additional kW of rooftop solar is associated with an average reduction of X kg CO₂/year, holding income constant”—useful for communication and policy. A black-box model might predict grid stability better but be hard to explain to regulators or the public.